

BTC Daily OHLCV → ML Features Guide

Reference guide for the btc_ml_features table generated from btc_price_history. The original OHLCV table remains unchanged. The target columns use next-day information and must NOT be used as input features.

1. Original OHLCV Columns

Column	Definition	ML Meaning / Use
date	Trading date of the daily BTC candle.	Time/index column; do not treat as a raw numeric price feature.
open	BTC price at the start of the daily candle.	Raw market feature.
high	Highest BTC price during the day.	Raw market feature.
low	Lowest BTC price during the day.	Raw market feature.
close	BTC price at the end of the daily candle.	Usually the main reference price.
volume	Total BTC trading volume reported by the data source for the day.	Raw volume feature; source-specific interpretation.

2. Return & Price-Movement Features

Column	Definition	ML Meaning / Use
return_1d	Percentage return over the previous 1 day: $\text{close}(t)/\text{close}(t-1)-1$.	Short-term momentum.
return_3d	Percentage return over the previous 3 days.	Short-term trend/momentum.
return_7d	Percentage return over the previous 7 days.	Weekly momentum.
return_14d	Percentage return over the previous 14 days.	Two-week momentum.
return_30d	Percentage return over the previous 30 days.	Monthly momentum.
price_change	Close minus Open for the same day.	Absolute daily candle movement.
price_change_pct	$(\text{Close} - \text{Open}) / \text{Open}$.	Daily percentage movement.

3. Candlestick Features

Column	Definition	ML Meaning / Use
candle_body	Close - Open.	Direction and size of the candle body.
candle_body_pct	$(\text{Close} - \text{Open}) / \text{Open}$.	Normalized candle-body size.
upper_wick	High minus the higher of Open and Close.	Measures rejection/price movement above the candle body.
lower_wick	Lower of Open and Close minus Low.	Measures rejection/price movement below the candle body.
upper_wick_pct	Upper wick divided by Close.	Normalized upper-wick size.
lower_wick_pct	Lower wick divided by Close.	Normalized lower-wick size.
daily_range	High - Low.	Total intraday price range.
daily_range_pct	$(\text{High} - \text{Low}) / \text{Close}$.	Normalized daily volatility/range.

4. Moving Averages

Column	Definition	ML Meaning / Use
sma_7	7-day Simple Moving Average of Close.	Short-term trend.
sma_14	14-day Simple Moving Average.	Short/medium-term trend.
sma_30	30-day Simple Moving Average.	Monthly-ish trend.
sma_50	50-day Simple Moving Average.	Medium-term trend.
sma_100	100-day Simple Moving Average.	Longer-term trend.
sma_200	200-day Simple Moving Average.	Long-term market trend.
ema_12	12-day Exponential Moving Average.	Faster-reacting trend measure.
ema_26	26-day Exponential Moving Average.	Slower trend measure used with MACD.
ema_50	50-day Exponential Moving Average.	Medium-term trend.

5. Momentum Features

Column	Definition	ML Meaning / Use
rsi_14	14-period Relative Strength Index, scaled 0–100.	Momentum/overbought-oversold context.
roc_7	7-day Rate of Change, expressed as percent.	Short-term momentum.
roc_14	14-day Rate of Change, expressed as percent.	Two-week momentum.

6. Volatility Features

Column	Definition	ML Meaning / Use
volatility_7	Standard deviation of daily returns over 7 days.	Recent volatility.
volatility_14	Standard deviation of daily returns over 14 days.	Two-week volatility.
volatility_30	Standard deviation of daily returns over 30 days.	Monthly volatility.

7. Volume Features

Column	Definition	ML Meaning / Use
volume_change	Percentage change in volume from the previous day.	Detects volume expansion/contraction.
volume_sma_7	7-day average trading volume.	Recent normal volume level.
volume_sma_30	30-day average trading volume.	Broader volume baseline.
volume_ratio	Current volume / 7-day average volume.	Shows whether today's volume is unusually high/low.

8. Trend-Ratio Features

Column	Definition	ML Meaning / Use
sma_7_30_ratio	SMA 7 divided by SMA 30.	Above 1 means short-term average is above the 30-day average.
sma_30_200_ratio	SMA 30 divided by SMA 200.	Compares medium-term trend with long-term trend.

9. Bollinger Band Features

Column	Definition	ML Meaning / Use
bb_middle	20-day moving average of Close.	Bollinger Band center.
bb_upper	20-day mean + $2 \times$ 20-day standard deviation.	Upper volatility band.
bb_lower	20-day mean - $2 \times$ 20-day standard deviation.	Lower volatility band.
bb_width	(Upper - Lower) / Middle.	Measures Bollinger Band expansion/contraction.
bb_position	(Close - Lower) / (Upper - Lower).	Relative position inside the bands; roughly 0 near lower band and 1 near upper band.

10. ATR Feature

Column	Definition	ML Meaning / Use
atr_14	14-day Average True Range based on true range.	Measures absolute recent price volatility; higher ATR means larger typical ranges.

11. MACD Features

Column	Definition	ML Meaning / Use
macd	EMA 12 - EMA 26.	Trend and momentum indicator.
macd_signal	9-period EMA of MACD.	Smoothed MACD reference line.
macd_hist	MACD - MACD Signal.	Difference between MACD and signal; momentum shift indicator.

12. Target Columns — NOT Input Features

Column	Definition	ML Meaning / Use
target_return	Next day's Close / today's Close - 1.	Regression target. This uses future information and must never be included in X.
target_direction	1 if next-day return > 0 , otherwise 0.	Optional classification target. Also must never be included in X.

How the ML Dataset Works

Input (X): OHLCV + engineered historical features.

Target (y): target_return for next-day regression OR target_direction for next-day classification.

Important: Every feature must be calculated using information available on or before that date. Do not use target_return or target_direction as model inputs. Use chronological train/validation/test splits rather than random shuffling.

Example

If BTC closes at \$100,000 today and \$102,000 tomorrow, today's target_return is 0.02 (+2%). The model sees today's historical features and tries to predict that next-day return.

Feature Groups at a Glance

Group	Purpose
OHLCV	Raw market price and volume information
Returns	Recent price momentum
Candles / Range	Intraday behavior and volatility
SMA / EMA	Trend direction and trend level
RSI / ROC	Momentum
Volatility	Recent variability of returns
Volume	Trading activity and unusual volume
Trend Ratios	Relationship between short/long trends
Bollinger Bands	Volatility and relative price position
ATR	Absolute market range/volatility
MACD	Trend + momentum
Targets	What the model is asked to predict